MARKETING SCIENCE
Vol. 8, No. 4, Fall 1989
Printed in U.S.A.

PRICE-INDUCED PATTERNS OF COMPETITION

ROBERT C. BLATTBERG AND KENNETH J. WISNIEWSKI
University of Chicago
A. C. Nielsen

This research focuses on how price changes influence the observed pattern of brand competition. The paper begins with a basic utility model formulation and examines the implications of three major classes of preference distributions on the expected patterns of competition. A price-tier model is proposed to operationalize the theory and to allow predictive testing. The price-tier model is estimated on 28 brands across four product categories.

The results show a specific asymmetric pattern of price competition. Higher-price, higher quality brands steal share from other brands in the same price-quality tier, as well as from brands in the tier below. However, lower-price, lower-quality brands take sales from their own tier and the tier below brands, but do *not* steal significant share from the tiers above. The results are consistent with a bimodal preference distribution, with the regular price indifference point being located toward the lower-quality end of the preference distribution for the categories analyzed.

(Promotional Analyses; Market Structure Models; Price Competition)

1. Introduction

How do brands within a retail category compete when one brand offers a price deal? What is the pattern of unit sales losses and gains? These issues have grown in practical and academic importance as temporary price reductions, or price deals, have become an important component of the manufacturer's and retailer's marketing strategy.

Promotional spending (including trade, sales force, and consumer) has grown dramatically since the mid-1970's, and currently accounts for over 65% of advertising/promotion expenditures in the U.S.[1] The 12% average yearly growth over the last decade is due in part to the fact that deals "work." Price deals can dramatically increase unit sales of the brand being promoted. However, exactly how and why price deals achieve the large promotional spikes typically seen in Universal Product Code (UPC) scanning data is not known. It is important for a retailer to understand how promotional prices affect brand competition, since brand switching affects the overall profitability of the retail category. In turn, the retailer's promotional profits affect the manufacturers' pass through of trade deals, and hence the overall profitability of their trade promotions.

This paper develops a basic framework for analyzing and explaining the pattern of deal price competition within a retail category. The primary focus is on the general characteristics of brand *price* competition within a product type and *not* on brand *attribute*

[1] Source: Donnelley Marketing (1986) based upon spending in retail channels by packaged goods manufacturers.

0732-2399/89/0804/0291$01.25

competition across product types. The theory deals with the short-run effects of price deals on demand, not on the long-run equilibria if such lower prices were maintained indefinitely. This article identifies how different promotional pricing structures affect the observed pattern of competition, given that brand attributes remain unchanged. The general shape of consumer preference distributions have important implications for which patterns of price competition will be observed. The concluding section comments on the implications of the research and on expanding the results to other types of product form competition.

2. Observations from the Literature

2.1. *The Literature on Price Promotions*

Few published studies are directly concerned with measuring the effects of price deals on unit sales, and still fewer examine how such price reductions affect the unit sales of other competitive brands in the retail line. Research on promotions tends to assume away competition (e.g. Sunoo and Lin 1978; Wilkinson, Mason, and Paksoy 1982), or assumes a specific pattern for the competitive effect (e.g. the implicit assumption of proportional share draw in the logit models of Guadagni and Little 1983 and McAlister 1984). There are several historical reasons for these limitations:

1. No theory exists to determine a priori which brands are likely to be highly price competitive within a product category.
2. The number of parameters in a series of brand level models increases significantly as competitive items are included, unless one assumes that all brands have the same price parameters.
3. Spurious correlations between one brand's prices and another brand's unit sales can occur when a large number of brands are simultaneously included in the estimation.
4. Historically, only limited (i.e. single brand) and/or poor quality sales and price data were readily available for nonproprietary research.

Assuming away competitive effects would not be serious if it could be shown that competitive dealing activities either did not significantly affect a given brand's sales or if competitive dealing activity affected all brands in some consistent way (e.g. the effect being proportional to market share). However, in at least three articles (Carpenter et al. 1988; Reibstein and Gatignon 1984; Batsell and Polking 1985), the effects seem to be asymmetric and unrelated to any specific summary statistic. Thus, the pattern of promotional price competition is not well understood.

2.2. *Other Approaches for Assessing Competitive Structure*

While much work exists on methodologies for identifying how brands may compete on the basis of product attributes and characteristics,[2] very little work exists on how *price* itself determines the observed pattern of competition. This is an important distinction. In hierarchial brand switching approaches, it may be that promotional prices or regular price changes, not simply "closeness" in terms of attributes, are causing the observed pattern of switching. For example, national brands and private label brands can have markedly different characteristics. If the national brands often price deal at substantial price discounts, a large amount of private label to national brand switching may result, implying a high degree of competitiveness between the classes of brands. If the national brands deal infrequently or deal at small discounts, little private label to national brand

[2] This substantial literature includes market structure approaches (see e.g. Kalwani and Morrison 1977; Urban, Johnson, and Hauser 1984; and Lilien and Kotler 1983 for examples and reviews); multidimensional scaling (e.g. Green and Wind 1973), and perceptual mapping and conjoint approaches (see e.g. Urban and Hauser 1980; Green and Srinivasan 1978; Johnson 1974; and Cattin and Wittink 1982).

switching may be observed, implying a low degree of competitiveness. Thus, it can be the regular or promotional price structure of the category, and not just comparability in attributes, that *causes* the observed switching.

In multidimensional scaling, perceptual mapping, and conjoint approaches, brands "close" to one another are viewed as competitive substitutes, regardless of the absolute level of price. Even if price is included as a dimension (e.g. as a feature in conjoint analysis or as a dimension in a perceptual map), the implication is that price competition should be symmetric or bi-directional. A price decrease for Brand A should result in a unit sales decrease for a "close" Brand B, and a price decrease for Brand B should result in a unit sales decrease for Brand A. As shown below, this implied symmetry of competition may be an artifact of the estimation technique used unless the underlying consumer preferences for the attribute(s) is uniform or unimodal.[3]

One innovation in perceptual mapping that resolves some of the above problems is DEFENDER (Hauser and Shugan 1983; Hauser and Gaskin 1984; Shugan 1987). DEFENDER explicitly incorporates price into the perceptual map, scaling product attributes in a "quantity of the attribute-per-dollar" measure (Hauser and Simmie 1981). However, DEFENDER implies that there exist at most 2 (or n) competitors for a given brand in a 2 (or n) dimensional analysis (n typically being 2 to 4). As shown later, though, it is possible for a large number of brands to be price competitive with one another to varying degrees, and for one brand to take share from another brand when it lowers its price, but not vice versa.[4]

One major concern with many of the approaches to assessing competitive structure is that they try to empirically derive the "best" structure for the market either by trying a multitude of possible structures or by a priori selecting specific structures based upon a variety of heuristics and then testing to see which fits best. As Rao (1984) noted, promotional price research in general "has sought predictions of the phenomena rather than cogent explanations." (p. S49). This criticism is especially valid as applied to research on how price changes themselves influence market structure. The remainder of this paper derives a basis for how price deals affect the nature of competition and analyzes how various consumer preference distributions may affect inter-brand competition. This paper seeks to explain why certain competitive patterns occur when price changes.

3. Model of Price Tiers

3.1. *A Basic Utility Theory of Price Competition*

Let consumer c's overall utility U_i^c be:

$$U_i^c = \theta^c q_i^c - p_i \tag{3.1}$$

where q_i^c is consumer c's perceived quality of brand i, p_i is the actual price, and $\theta^c > 0$ is consumer c's willingness to pay for quality (i.e. θ^c is the importance weight on overall quality relative to an importance weight of 1 on price p_i).[5] Quality is defined here as a summary measure denoting the brand's overall attractiveness, exclusive of price. As such, quality is an overall preference for a particular usage occasion that summarizes

[3] It may be possible to extend some of these approaches to incorporate asymmetry by including interaction effects. Note too that the above approaches tend to use regular prices rather than deal prices.

[4] DEFENDER can handle different absolute amounts of sales loss depending on the underlying assumptions of the consumer preference distribution across product attributes. For an additional competitor to become efficient (significant) requires driving unit sales for one of the adjacent (close) competitors to zero unit sales. (See e.g. Hauser and Gaskin 1983.)

[5] This basic model is somewhat comparable to the approach taken in the self-selection literature in economics. See e.g. Mussa and Rosen (1978), Cooper (1984), Moorthy (1984), and Oren, Smith and Wilson (1984). However, equation (3.1) deals with quality as a perceived versus an objective dimension.

multidimensional consumer product perceptions or attributes, exclusive of price. Note that some consumers may place much greater importance on quality (have a higher θ) than others, and that this idiosyncratic desire for quality can have a significant effect on the price a given consumer is willing to pay for a specific level of quality.[6]

For the consumer comparing two brands i and k, the difference in utility is:

$$U_i^c - U_k^c = (\theta q_i^c - p_i) - (\theta q_k^c - p_k) = \theta^c(q_i^c - q_k^c) - (p_i - p_k). \tag{3.2}$$

Thus, given that both prices are below the consumer's reservation price, the decision rule becomes:

Choose brand i if:

$$\theta^c(q_i^c - q_k^c) > (p_i - p_k), \qquad \text{or} \tag{3.3a}$$

Choose brand k if:

$$\theta^c(q_i^c - q_k^c) < (p_i - p_k). \tag{3.3b}$$

Suppose $(q_i^c - q_k^c) > 0$ and equation (3.3b) describes a particular consumer. Brand i is superior in quality to brand k, but this consumer is currently a buyer of brand k. As $p_i - p_k$, eventually the difference $(p_i - p_k)$ will become small enough for equation (3.3a) to hold since $\theta^c(q_i^c - q_k^c) > 0$, and this consumer will switch from brand k to brand i.

Suppose next that another consumer is currently described by equation (3.3a). At current prices, the consumer buys the higher quality brand i. As p_k declines, the consumer will buy brand k only if $\theta^c(q_i^c - q_k^c)$ is sufficiently small. In words, this means the consumer will switch down to the lower quality brand only if the perceived quality difference between the two brands is small and/or his strength of preference for quality relative to price is small. Even if $p_k = 0$, that is, brand k is free, it is still possible that the consumer will not shift to brand k if $\theta^c(q_i^c - q_k^c) > p_i$. Thus, depending on the distribution of preferences in the population and prices, some consumers (the higher θ's) may never be affected by price promotions of lower quality brands, while others may switch back and forth between products depending on the relative prices $(p_i - p_k)$ and their (lower) value of θ^c.

3.2 *Population Preference Distributions*

For ease of terminology, define R^c as consumer c's *relative preference* for brand i relative to brand k. Thus, let $R^c = \theta^c(q_i^c - q_k^c)$. If $q_i^c > q_k^c$, a consumer's relative preference for brand i increases as the perceived overall quality difference $(q_i^c - q_k^c)$ increases and/or as the importance consumer c places on quality θ^c increases. The pattern of brand competition in a market is thus a function of the shape of the distribution of relative preferences R^c over all consumers c. Little is known empirically about the shape of this relative preference distribution or its underlying components, the distributions of θ^c and $(q_i^c - q_k^c)$.

Three general distributions shown in Figures 1 and 2 can be examined to illustrate the range of the possible effects that the shape of the relative preference distribution R can have on the pattern of price competition. One assumption is a uniform distribution. A second is a normal distribution, assuming that an individual's relative preference is the result of many influences of the buyer's environment and thus is normally distributed because of the Central Limit Theorem. The normal distribution is an example of a

[6] As θ^c is larger, indicating a stronger desire for quality, price p_i can be larger for a given quality level q_i and still leave U_i^c unchanged. Thus, as the value of θ^c increases, the consumer is willing to pay more for the same absolute level of perceived quality.

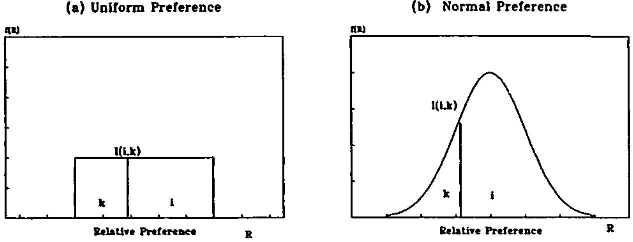

FIGURE 1. Price Competition Symmetric in Direction

unimodal distribution, while the uniform distribution is a constant. A third distribution is a "U-shaped" or a "J-shaped" distribution (such as a Beta distribution) as shown in Figure 2. Thus, the distributions in Figures 1 and 2 represent the proportion of consumers $f(R)$ with a specific value for the relative preference $R^c = \theta^c(q_i^c - q_k^c)$. For example, Figure 2a shows the relative preference distribution to be bimodal, with a large proportion of the population having low values of R (i.e. viewing the brands as having small quality differences and/or having a lower willingness to pay for quality) and a smaller proportion of the population with a large value R for their relative preference for brand i relative to brand k.

Equation (3.1) implies that a consumer will be indifferent between either of a pair of brands if:

$$\theta^c(q_i^c - q_k^c) = R^* = p_i - p_k = I_{ik}. \tag{3.4}$$

The vertical lines in Figures 1 and 2 represent the numerical value R^* at the point of indifference I_{ik}. At point I_{ik}, prices of brands i and k are such that a consumer receives the same level of overall utility if he/she buys either brand i or k. Consumers to the left of I_{ik} prefer brand k, since for them $\theta^c(q_i^c - q_k^c) < (p_i - p_k)$, while consumers to the right of I_{ik} prefer brand i. As brand i reduces its price, the point of indifference in Figure 1 and 2 moves to the left, and additional buyers of k are now willing to switch up to a purchase of i, even though $p_i > p_k$. The narrowing of the price difference is enough to cause some consumers to switch to the higher quality, yet still higher priced brand. This is caused by their willingness to pay for quality θ^c or the difference in perceived quality between the brands $(q_i^c - q_k^c)$ being high enough to cause the overall utility for brand i to become greater than the overall utility for brand k, given the reduction in price p_i [(i.e., $\theta^c(q_i^c - q_k^c) = R^* > p_i - p_k$)]. Thus, brand k loses customers, and brand i increases share.

In Figure 1, for the uniform and normal distributions, price competition operates in both directions in the sense that a price reduction in one of the brands results in significant numbers of consumer switching from the nondealt to the dealt brand. In other words,

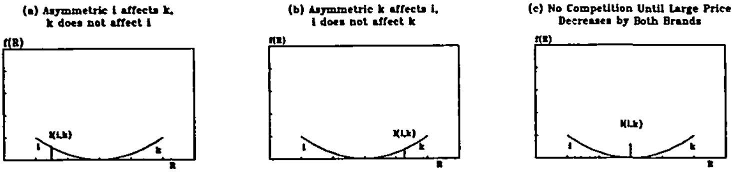

FIGURE 2. Asymmetric and Independent Price Competition.

cross-price elasticities can be expected to be positive and significant for both pairs of brands.[7]

Figure 2 helps explain a specific type of observed asymmetry in price competition. Figure 2a shows the case where price decreases by brand i take significant (nonzero) unit sales from brand k, but as brand k reduces its price, negligible sales loss occurs for brand i. Figure 2b reverses the order of this effect, and Figure 2c shows the case where the distribution of relative preferences is so polar between the brands that negligible sales loss occurs for either brand as the other price deals until large price cuts are in effect.

Note that the above discussion is based upon a paired comparison of those buyers who currently prefer either brand i or brand k to all other brands under "normal" or regular prices. There is no difficulty in extending the discussion to three brands or more, since the relevant comparison will *always* be between *two* brands. In equilibrium, the market consists of buyers of i, j, k, l, etc., each consumer purchasing the brand which generates the greatest utility according to equation (3.1). The above discussion is centered on what will happen to a given brand's baseline unit sales when a competitive brand reduces its price, and vice versa.

If brands i and j are high quality brands, and brand k is a low quality brand, it is possible that the $i-k$ and $j-k$ relative preference distributions may look like 2a, while the $i-j$ distribution may look like 1a or 1b. (Recall that while the willingness to pay for quality parameter θ^c is a constant for a given consumer across all brands, the quality of the brands q_i^c varies across the brands i, j, k, etc.) Thus, any given brand may compete with any number of competitive brands.[8]

3.3. *Theory of Price Tiers*

In many markets, one sees groups of brands with different prices and implied quality. For example, Sears offers their customers three grades at three price levels: good, better, best. In packaged goods there are national, private label and generic brands. Table 1 shows regular prices[9] offered for four grocery product categories over time for one grocery chain in the Chicagoland market. It shows national, moderate/private label and generic brands having significant price differentials. These separate distinct retail price groups will be called *price tiers*.

With a few relatively weak and plausible assumptions, it is possible to develop specific predictions about the structure of price competition between and within price tiers.

Assumption (1). In general, national brands (N) have higher perceived quality levels than moderate/private label brands (P); private labels (P) are higher quality than generics (G). Thus, $q_N^c > q_P^c > q_G^c$.

Based on this assumption, on average $\theta^c(q_i^c - q_k^c) > 0$ for i being a national brand and k a private label, or for i being a private label and k a generic brand. Different consumers

[7] Note that these cross-price elasticities need *not* be comparable in magnitude. In fact, if brands i and k are identically priced, Figures 1a and 1b imply the cross-price elasticity of i's effect on k is larger than the effect of k on i, simply because i's market share is greater. This is because for a fixed distance movement (fixed percentage change) of the indifference point I, brand k loses proportionately more of its base customers than does brand i, and hence its cross-price elasticity with k is greater than k's cross-price elasticity with i.

[8] Figure 2 can also be used to explain how DEFENDER can accomodate a form of asymmetric competition. In DEFENDER terminology (see e.g. Hauser and Shugan 1983, Shugan 1987), if the horizontal axis R in Figure 2 is viewed as the angle of the consumer's linear indifference curve with the vertical-axis attribute, then such a U-shaped distribution of preferences (angles) across attributes can generate asymmetric competitive effects of the type described above. A limitation, though, is that DEFENDER will only allow adjacent brands to compete with a given brand, limiting competition to at most n competitors in n-dimensional attribute space. In addition, such an approach requires a very complex pattern of indifference curves (preferences) in n-dimensional space (i.e. what do U-shaped preferences mean in 4-dimensional space) versus the simpler univariate overall relative preference approach described in the text.

[9] Regular prices are the nondeal regular shelf prices of the brands (i.e. the prices facing the consumer during nonpromoted weeks).

TABLE 1

Price Tiers—Brands and Regular Price Ranges (Brands Ordered by Market Share Within Each Tier)

Flour (5 lb. All Purpose)		Margarine (1 lb. Sticks)		Bathroom Tissue (4-Roll Equivalent)		Tuna (6.5 Oz. in Oil)	
PREMIUM BRANDS							
Pillsbury	.98–1.23*	Imperial	.89	Charmin	1.39	Bumble Bee	1.09–1.19
Gold Medal	.98–1.19	Land O'Lakes	.75–.83	Northern	1.39	Chkn of Sea	1.09–1.37
Ceresota	.98–1.19	Parkay	.83	White Cloud	1.39	Starkist	1.09–1.37
		Bluebonnet	.85	Soft 'N Pretty	1.39		
		Chiffon	.59–.69				
MODERATE BRANDS							
Jewelmaid	.89–.99	Bluebrook	.49–.53	Scott Tissue	1.06[1]	Bluebook	.89–1.05
		Sunnyland	.49–.53	Coronet	.995[2]		
				Jewel	.99		
				Sable Soft	1.11		
GENERIC BRANDS							
Generic	.69	Generic	.35	Generic	.68–.69	Generic	.73

* Represents the actual regular price range observed throughout the data history.

[1] Scott Tissue is sold as 53¢/100-sheet roll; 2–3 rolls of Scott yield approximately 4 rolls of the premium brands on a volume basis.

[2] Coronet is sold in an 8-pack for $1.99, or .995 for 4 rolls.

may have different perceived quality for the brands, but for the purposes of the empirical analyses q_i^c is generally assumed to be greater than q_k.

Assumption (2). There is a positive correlation between price and quality. Thus $p_N > p_P > p_G$.

In general, retailers such as Sears, Roebuck and Co. and Jewel Food Stores and most manufacturers seem to set pricing policy based on the observation that increased levels of product quality for a particular type of product also produce increased costs and hence increased prices. Gabor and Granger (1966) find strong evidence that consumers do use price as an indicator of quality at the individual level.

If one accepts these assumptions, then predictions about brand price competition can be made. The predictions depend on the shape of the distribution of the consumers relative preferences R^c and on the location for the regular price indifference point I_{ik}.

Assuming *comparable* levels of quality q_i and q_k and comparable retail prices, competition *within* a price/quality tier should be bi-directional and resemble one of the situations shown in Figure 1. Thus, each brand significantly affects the other's unit sales when they price deal, although the absolute magnitude of sales loss need not be equal.

The most interesting predictions are those comparable to Figures 2a and 2b. What follows is a discussion of 2a, although the reverse argument can be made for case 2b. If Figure 2a describes the market's relative preferences, then when a brand i in a higher price tier reduces its price, it attracts buyers from the lower price tier brand k. Graphically, the indifference point moves to the left, and consumers who now find themselves to the right of the indifference point (i.e. R^c is now greater than I_{ik}) now buy brand i. However, when a brand in the lower tier reduces its price, brands in higher tiers lose little volume, since there are few consumers with relative preferences immediately to the right of the indifference point I_{ik}. (Reasons why this situation may occur are discussed in §7.) Within tiers, brands affect other same-tier brands as they price deal, since the perceived quality and retail price differences are assumed to be much smaller within than between tiers, and the shape of the distribution of relative preferences is more likely to look like Figures 1a or 1b.

How to distinguish between 1b and 2a using aggregate data is an empirical issue. Both result in a form of asymmetric or unidirectional competition. However, note that in general we would expect case 1b to show substantial percentage losses in brand i's market share, implying large sales losses for brand i during brand k's dealing periods relative to the usual fluctuation of brand i's baseline sales. Thus, we would expect to observe significant cross-price effects in case 1b. However, for case 2a, the loss in i's sales relative to its market share or baseline sales is minimal. Thus, sales loss relative to regular sales fluctuation should be minimal, and the cross-price coefficient will be small. The empirical statistical results reported later indicate that this is a logical way to distinguish between 1b and 2a.

§5 tests these predictions with the data described in the next section.

4. Description of the Data

To evaluate the theory described in the last section, data from four product categories were available at the time of this analysis: all-purpose flour, stick margarine, bathroom tissue, and chunk light tuna in oil. Forty-nine weeks of data were supplied by Jewel Food Stores, the leading market share grocery chain in Chicago. The data collected for this study include unit sales by item, price by item, newspaper feature advertising, and price deals by item. The dependent variable of interest is total unit sales across 26 stores in the sample. The 26 stores all have the same retail prices, and the timing of price changes and deals within these stores is identical.

Because this study's focus is on price and not on attribute effects, brands were selected to be comparable on the basic attributes. Thus, only regular stick margarine brands were studied in the margarine category. Omitted were tubs of margarine, corn oil margarine, and squeeze margarine. Similarly in tuna, only 6.5-oz. light chunk tuna in oil was studied. This reduced other nonprice effects by making the defined category more homogeneous.

Several issues arose in creating the data set. First, a hardware problem at the retailer resulted in the destruction of the data for week 17, reducing the usable sample to 48 weeks. Second, the original private label bathroom tissue was phased out during weeks 39–49. Estimations on this one brand use only the "stable" first 38 weeks of data. The third problem is common to many retailer scanning systems. Weekly data were collected Monday through Sunday and the price reported on the data tape was the Sunday price. In the Chicago SMSA, price deals are typically offered Thursday through Wednesday. If the price was higher (or lower) during the first part of the week (e.g. Monday–Wednesday), this was not recorded. Thus, it is necessary to modify some of the post-deal-period prices to reflect the actual price facing the consumer in a particular week. Because data on advertised price deals were collected, the deal length was known and the prices modified accordingly. Thus, if a deal was available during any part of the week, the price for that week was set to the deal price. This requires a variable in the model to adjust for the percentage of a given week in which the deal was offered, as noted in equation (5.1), §5.

5. Model Used for Testing Predictions

The structure of price competition is identified by examining the patterns of competitive brands' cross-price effects. The purpose of this section is to present a model that allows a test of the distributional assumptions and the price tier theory. The model used for estimation purposes is shown in equation (5.1).

The model is a modification of functional forms used by the authors in other promotional scanning research. It incorporates nonprice variables such as newspaper feature advertising, deal decay (which allows an exponential decay in a promotion's effectiveness), seasonality, and a data period adjustment to handle the fact that data are supplied on a Monday–Sunday scanning week while promotions in the Chicago market generally run Thursday through Wednesday. These nonprice variables which are not the focus of this paper allow more precise estimation of the price effects. The model incorporates three price variables to estimate own-brand regular price, own brand deal price, and competitive brand effects.

The own-price elasticity is $\beta_1 P_{it}$. The own-brand deal discount is modeled using a percentage discount from regular price, $[(P_{it} - d_{it})/P_{it}]$, and the elasticity is $\beta_2 d_{it}/P_{it}$. This implies consumers are evaluating percentage changes in deals, and is consistent with recent work which implies consumers frame the concept of a "good deal" on a relative instead of an absolute basis (e.g. Thaler 1985). Tests of other variable specifications (e.g. absolute penny discounts instead of the percentage discount used here) show that the percentage discount from regular price works as well or better (based on an R-squared measure of fit) than alternative specifications for over 80% of the brands analyzed, with the percentage decrease specification generally a close second (in R-Squared terms) in the other 20% of the cases (Wisniewski and Blattberg 1988).

The competitive price effects elasticities, which are the key focus of this study, are $(-\gamma_{ij}/d_{jt})$. The elasticities are point elasticities computed at the last regular price observed in the data history. Although using the mean regular price or some other measure of regular price changes the point estimates, the *pattern* of results is identical to that reported below.

The purpose of this research is to analyze the *pattern* of price competition in the market. Other functional forms can be specified, and several other forms were also estimated in this and other research (see e.g. Wisniewski and Blattberg 1988; Allenby 1988).

$$S_{it} = \exp(c_i - \beta_1 P_{it} + \beta_2 (P_{it} - d_{it})/P_{it} - \sum_{j \neq i} \gamma_{ij}/d_{jt} + \alpha_1 A_{it} - \lambda_1 T_{it} + \alpha_2 W_{it} + \alpha_3 \phi_i) \quad (5.1)$$

where

S_{it} = total unit sales of brand i in period t (aggregated over the 26 stores in the price zone)
P_{it} = regular price of brand i at time t,
d_{it} = actual price of brand i at time t,
A_{it} = 1 if brand i advertised at time t, 0 otherwise,

$$T_{it} = \begin{cases} \textit{deal decay variable}: \\ 0 \text{ during nondeal weeks}, \\ n-1 \text{ during the } n\text{th week of a multi-week deal}, \end{cases}$$

scanned versus deal week adjustment variable:

$$W_{it} = \begin{cases} 0 \text{ during nondeal weeks}, \\ -.43 \text{ during the first week of a deal (i.e. where the deal price is in effect from Thursday–Sunday)},^{10} \\ 0 \text{ during all "full-week" deal weeks (i.e. all deal weeks where the deal price is in effect all 7 days)}, \\ -1.05 \text{ during the last week of a deal (i.e. all deal ending-period weeks where the deal price is in effect only Monday–Wednesday of the week)}, \end{cases}$$

ϕ_{it} = 1 during a "seasonal" period for the category, 0 otherwise (seasonal periods included Easter, Thanksgiving, and Christmas weeks),
c_i = constant for brand i,
α_i = feature advertising parameter,

[10] $e^{-0.43} = 0.65$ and $e^{-1.05} = 0.35$, the adjustments used for deals. In the current data base approximately 65% of the 1-week promotions effects occur during the Thursday–Sunday period, and some 35% occurs during the Monday–Wednesday period.

λ_i = exponential deal decay parameter,
β_1 = own brand price effect,
β_2 = own deal price effect,
γ_i = competitive brand price and deal effects.

However, the results in terms of the symmetric/asymmetric nature of promotional price competition are virtually identical. Thus, §6 presents the results of estimating equation (5.1) on the four product categories.

6. Analyses of Intra-Category, Interbrand Price Competition

The model shown in equation (5.1) was estimated for 28 brands: five 5-lb. all purpose flours, eight 1-lb. regular stick margarines, ten bathroom tissues, and five 6.5 oz. size tuna fish in oil brands. The estimation was performed over all 48 weeks of existing data. Since the objective of the estimation is to identify the structure of price competition, no holdout sample was used. This maximized the degrees of freedom available. The estimation technique used was OLS regression. Analysis of the residuals shows no significant correlations among the residuals between brands in the same category beyond the number expected by chance. Thus, use of SUR (seemingly unrelated regression) offers little additional information.

The summary of the estimation results for the relevant cross-price terms is presented in Table 2. The regular retail price at the end of the data history is used to compute the elasticities. Whether this last price or the average price is used makes no difference for the following discussion. Adjusted R^2 is in the range of 0.75 to 0.95, except for generics which are poorly modeled by any of the price and advertising variables in the data base. The margarine category, which did not have any special display activity associated with its deals at the time of this analysis, generally has the best fit, though bathroom tissue, having substantial display activity, is almost as high.

Figure 3 presents typical fitted versus actual sales histories for the largest-share premium brands in each category. For all brands except the generics the model given in equation (5.1) accurately fits the peaks and the decay patterns for price deals. This is an indication that the major competitive factors have been captured by the estimation model. Even in nondeal periods, the models do an adequate job of tracking the upward and downward movements in the data histories, indicating that competitive prices are affecting week-to-week unit sales.

6.1. *Summary of the Cross-Price Results*

There are 171 price coefficients estimated across these 28 brands, 135 being cross-price terms and hence particularly appropriate for this analysis. (Eight cross-price coefficients could not be estimated due to lack of price variation.) Since the generic brands did not often significantly reduce price, a separate 0/1 variable[11] was used to try to assess the effect of generic deals on other brands. Table 3 summarizes the significance of the coefficients based on a significance level of 0.05 ($t = 1.697$, $n = 30$), and also reports average price elasticities. Table 3 counts the effects as follows: For a given brand i, if any other brand's price within a tier affects the brand i, then the other brand's price-tier is said to affect the price tier of the brand i. For example, in margarine, there are 5 national brands, 2 private labels, and 1 generic. For a specific national brand, if any of the other 4 national brands' prices affects the specific brand, Table 3 says the national brand is affected by other national brands' prices. Since there are 5 national brands, there are 5 relevant national brand comparisons. Similarly, if *either* of the two private labels affects a given national brand's sales, then it would be counted as a moderate tier brand affecting a national brand's unit sales.

[11] A 0/1 generic-deal indicator variable was used to denote the absence/presence of a generic deal.

TABLE 2

Cross-Price Elasticities

FLOUR	AFFECT UNIT SALES OF:				
PRICES OF:	PIL	GDM	CER	JM	GNC
Pillsbury (PIL)	—	1.15	0.67	2.44	0.65
Gold Medal (GDM)	0.99	—	1.21	1.18[a]	0.01[a]
Ceresota (CER)-Deal	(0.06)*	0.03*		0.37*	0.03*
Jewel Maid (JM)	0.19[a]	1.74	0.08[a]	—	0.43[a]
Generic (GNC)-Deal	0.01*	0.03*	0.01*	0.15[a]	—
Market Share	0.37	0.09	0.11	0.08	0.34
Adjusted R^2	0.96	0.89	0.94	0.77	0.24

MARGARINE	AFFECT UNIT SALES OF:							
PRICES OF:	IMP	LOL	PKY	BBT	CHF	BBK	SNY	GNC
Imperial (IMP)	—	0.84	0.71	0.27[a]	0.22*	0.00*	0.64	0.03*
Land O'Lakes (LOL)	0.16*	—	0.63	0.50	0.44	0.05*	0.75	0.06*
Parkay (PKY)	0.13*	0.34	—	0.32	0.23	0.15	0.20[a]	(0.04)*
Bluebonnet (BBT)	0.49	0.23	0.25	—	0.43	0.10*	(0.14)*	(0.08)*
Chiffon (CHF)	(0.14)*	0.19*	0.37*	0.43*	—	(0.24)*	0.85[a]	0.01*
BlueBrook (BBK)	(0.20)*	(0.13)*	0.05*	0.06*	(0.14)*	—	0.30[a]	0.19
Sunnyland (SNY)	0.03*	0.21[a]	0.19*	0.27[a]	0.02*	0.25	—	0.13
Generic (GNC)-Deal	0.01*	(0.16)	(0.06)*	(0.05)*	0.00*	0.01*	0.10*	—
Market Share	0.17	0.14	0.12	0.06	0.04	0.32	0.08	0.07
Adjusted R^2	0.91	0.95	0.96	0.95	0.88	0.93	0.88	0.66

BATHROOM TISSUE	AFFECT UNIT SALES OF:								
PRICES OF:	CHM	NTH	WTC	SNP	SCT	COR	JWL	SBL	GNC
Charmin (CHM)	—	1.31	1.47	1.83	0.60*	(0.11)*	0.81*	1.33	(0.37)*
Northern (NTH)	0.80	—	0.72	1.58	0.18*	(0.24)*	0.80	1.56	0.39*
White Cloud (WTC)	1.14	0.90[a]	—	1.69	0.22*	0.44*	0.70*	0.98	(0.40)*
Soft 'N Pretty (SNP)	0.74	0.89	0.56	—	1.25	0.02*	0.71[a]	1.75	(0.90)
Scottissue (SCT)	0.27*	0.20*	0.26*	1.37[a]	—	1.50	1.99	1.30	0.92[a]
Coronet (COR)	0.24	0.02*	0.05*	0.22*	0.34[a]	—	1.04	0.39	0.64
Jewel (JWL)	0.19[a]	(0.40)*	0.00*	0.96[a]	0.24*	1.92	—	0.17	0.18
Sable Soft (SBL)	ND	ND	ND	ND	ND	ND	ND	ND	ND
Generic (GNC)	0.04*	(0.12)*	0.01*	(0.06)*	0.00*	(0.10)*	0.15[a]	0.03*	—
Market Share	0.15	0.10	0.07	0.05	0.12	0.03	0.03	0.03	0.42
Adjusted R^2	0.84	0.90	0.91	0.86	0.76	0.71	0.88	0.64	0.17

TUNA FISH	AFFECT UNIT SALES OF:				
PRICES OF:	BBB	CKN	STK	BBK	GNC
Bumble Bee (BBB)	—	0.64	0.37*	1.55	0.10*
Chicken/Sea (CKN)	1.05[a]	—	1.26	0.25*	0.20*
Starkist (STK)	0.85[a]	0.09[a]	—	0.54[a]	(0.27)[a]
Blue Brook (BBK)	0.36*	0.07*	0.17*	—	0.31*
Generic (GNC)	ND	ND	ND	ND	ND
Market Share	0.25	0.21	0.16	0.07	0.31
Adjusted R^2	0.84	0.90	0.88	0.69	0.04

All cross-elasticities significant at the 0.05 level unless otherwise noted.

[a] Significant at the 0.10 level.

* Insignificant at the 0.05 level.

( ) Indicates an incorrectly signed coefficient.

Deal—Some of these brands did not extensively price deal. The value reported is the coefficient of a 0/1 advertising variable for that brand's deal.

ND—No deals of any type for this brand.

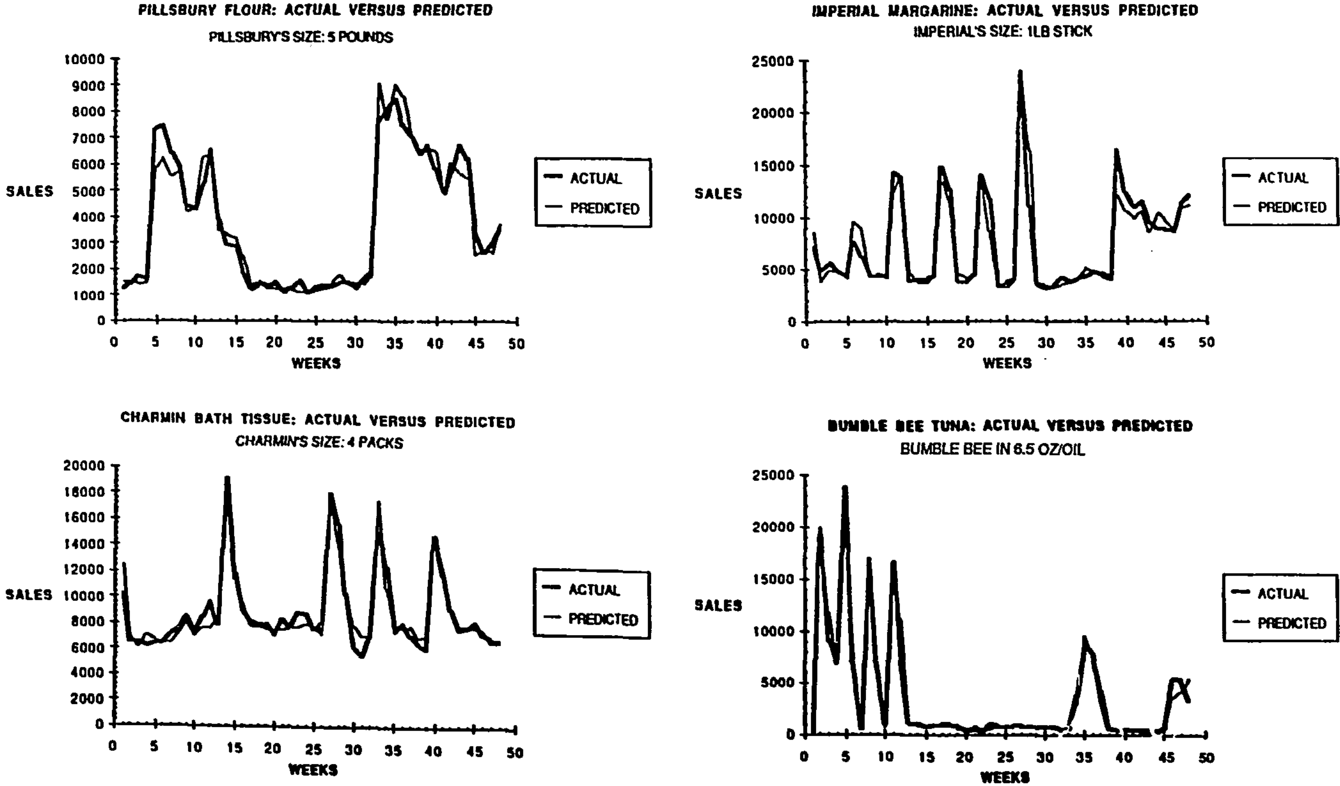
PILLSBURY FLOUR: ACTUAL VERSUS PREDICTED
PILLSBURY'S SIZE: 5 POUNDS
SALES
10000
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0 5 10 15 20 25 30 35 40 45 50
WEEKS
ACTUAL
PREDICTED
IMPERIAL MARGARINE: ACTUAL VERSUS PREDICTED
IMPERIAL'S SIZE: 1LB STICK
SALES
25000
20000
15000
10000
5000
0
0 5 10 15 20 25 30 35 40 45 50
WEEKS
ACTUAL
PREDICTED
CHARMIN BATH TISSUE: ACTUAL VERSUS PREDICTED
CHARMIN'S SIZE: 4 PACKS
SALES
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0 5 10 15 20 25 30 35 40 45 50
WEEKS
ACTUAL
PREDICTED
BUMBLE BEE TUNA: ACTUAL VERSUS PREDICTED
BUMBLE BEE IN 6.5 OZ/OIL
SALES
25000
20000
15000
10000
5000
0
0 5 10 15 20 25 30 35 40 45 50
WEEKS
ACTUAL
PREDICTED

FIGURE 3.

TABLE 3

Summary of Predictive Hypothesis Results and Average Cross-Price Elasticities

PRICE CHANGES OF:	AFFECT UNIT SALES OF: PREMIUM				MODERATE				GENERIC			
				$\bar{\gamma}$				$\bar{\gamma}$				$\bar{\gamma}$
PREMIUM	FLOUR	3/3*	14/15	0.69	FLOUR	1/1	7/8	1.33	FLOUR	1/1	1/4	0.23
	MARG.	5/5		0.35	MARG.	2/2		0.24	MARG.	0/1		0.00
	TISSUE**	4/4		1.14	TISSUE	3/4		0.69	TISSUE	0/1		−0.32
	TUNA	2/3***		0.71	TUNA	1/1		0.78	TUNA	0/1		0.01
				$\bar{\gamma}$				$\bar{\gamma}$				$\bar{\gamma}$
MODERATE	FLOUR	1/3[a]	2/15	0.67	FLOUR	NA	4/6	—	FLOUR	0/1	2/4	0.43
	MARG.	0/5		0.04	MARG.	1/2[b]		0.28	MARG.	1/1		0.16
	TISSUE	1/4[c]		0.33	TISSUE	3/4		0.99	TISSUE	1/1		0.58
	TUNA	0/3		0.20	TUNA	NA		—	TUNA	0/1		0.31
				$\bar{\gamma}$				$\bar{\gamma}$				$\bar{\gamma}$
GENERIC	FLOUR	0/3	1/12	0.02	FLOUR	0/1	0/7	0.15	FLOUR	NA	NA	—
	MARG.	1/5[d]		0.01	MARG.	0/2		0.06	MARG.	NA		—
	TISSUE	0/4		−0.03	TISSUE	0/4		0.02	TISSUE	NA		—
	TUNA	NA		—	TUNA	NA		—	TUNA	NA		—

* Read as: "In 3 (numerator) out of 3 (denominator) cases, at least 1 of the premium brands (row type) prices affect at least 1 of the premium brands (column type) unit sales" ($\alpha = 0.05$, $t = 1.645$).

** Tissue category excludes Waldorf—see text.

*** 3/3 at the 0.10 level.

[a] Jewelmaid → Gold Medal.

[b] Bluebrook → Sunnyland (Bluebrook → Sunnyland at 0.10 level).

[c] Coronet → Charmin.

[d] Generic → Land O'Lakes (coefficient has wrong sign).

NA Not Applicable/Only 1 brand, no comparison possible.

$\bar{\gamma}$ Average cross-price elasticity or average deal cross-elasticity.

Thus, Table 3 presents a "worst-case" analysis which favors finding *symmetric* or bidirectional competitive price effects. Overall, the results strongly support *asymmetric* or unidirectional price tier competition of a very specific type.

We discuss first the diagonal and cells in the triangle above the diagonal. In 14 of 15 comparisons (15 of 15 at the 0.10 level), premium priced brands steal significant unit sales from other premium priced brands when they price deal. In 7 of 8 cases, premium priced brands take sales away from brands in the moderate or private label category when premium prices are reduced. In 1 of 4 cases (Pillsbury flour) a premium brand's price affects the generic brand. In 4 of 6 cases (5 of 6 at the .10 level), moderate/private label brand prices affect unit sales of other moderate/private label brands, the only exception (.10 level) being Scottissue, discussed below. In 2 of 4 cases, the moderate/private label brands also exhibit statistically significant unit sales effects on the generic brands. Thus when higher-price, higher-quality brands price deal, they steal sales from their own tier and the tier below.

In Table 3 the three cells below the diagonal provide the strongest evidence of asymmetry in the direction of competition. In only 3 of 32 comparisons do lower-tier brands statistically significantly affect upper-tier brand unit sales. The results are even stronger than they seem at first. In 2 of 4 categories, there are multiple moderate/private label brands (2 private label margarines, 4 moderate brand bathroom tissues). If any one of these moderate brands affects the premium brand, it is counted as a "violation" in Table 3. In fact, there are 45 moderate and lower tier coefficients summarized in the 3 cells below the diagonal. In only 3 of the 45 cases (6.7%) are the coefficients statistically significant.

Note that 2.25 (or 4.5) would be expected simply by chance at the 0.05 (0.10) level. One of the significant coefficients (the effect of a generic deal on Land 'O Lakes margarine) is actually positive (indicating that in the periods when the generic brand advertised, Land 'O Lakes unit sales were *up*), and is likely to be a statistical artifact.

The above results are also consistent with the absolute magnitudes of the cross-price elasticities. The average elasticities reported in Table 3 show strong own-tier effects, and premium/private label cross-elasticities that are greater than their private label/premium counterparts. The absolute magnitude of the private label's cross-price elasticities with the national brands is almost always much smaller than the national brand's cross-elasticities with the private labels. As Tables 2 and 3 indicate, moderate-affecting-premium brand cross-elasticities are typically smaller, often are insignificant, and sometimes are even negative.

The other two below-diagonal significant coefficients are instructive. Bimodal preference distributions do not rule out a lower-tier brand affecting an upper-tier brand. Such a result suggests that either the differences in quality are closer than for other across-tier pairs of brands, or the price of the lower-tier brand is being reduced sufficiently to offset the loss in utility from switching from the higher-tier brand to the lower-tier brand for some of the consumers in the market. In one of these cases, the moderate-tier private label flour Jewelmaid draws share from the upper-tier brand Gold Medal. Gold Medal flour is a national brand, priced at nearly the same price of the market leader Pillsbury. Yet, Pillsbury enjoys approximately a 37% share of the market, while Gold Medal languishes at 9%. Thus, there would seem to be some perceived differences in the mean level of quality offered by the two brands although prices are comparable. The results imply that the difference in perceived quality between Jewelmaid and Gold Medal is significantly less than the difference in the quality between Pillsbury and Jewelmaid.

This same argument cannot be made for the moderate brand Coronet bathroom tissue affecting the market share leader Charmin. All premium brands of bathroom tissue were regularly priced at $1.39 for four rolls. Of these 4 premium brands, Charmin was the market share leader. Coronet was the *only* 8-roll pack in our analysis (priced at $1.99, hence the highest *absolute* price). Coronet also is the only moderate brand *not* affected by other moderate brands' pricing. Thus, we are unable to determine if this effect is real, if it is due to a high absolute price causing consumers to perceive it to be in a higher price tier, or if it is simply a chance occurrence.

All of the other 25 brands are statistically consistent with price-tier theory of the type shown in Figure 2a or 2c. This means that "upper-tier" brands tend to affect own tier and tier below brands when they deal, but lower tier brands do not affect the upper-tier brands significantly. Occasionally, there may be no statistically observable effect between or within tiers.

Two brands, Scottissue and Chiffon, illustrate the value of price-tier theory in assessing competitive market structure. Based on square-footage, it was initially unclear whether Scottissue was a premium (3 rolls for $1.53) or a moderate (2 rolls for $1.06) brand. Scott's advertising strategy seems to focus on economy as its core benefit, implying it is positioned as a moderate brand. Its regular price is 53¢ per roll, the lowest absolute price in the category. The estimation results imply that consumers treat Scott as a moderate rather than a premium priced brand. It does *not* affect other national brands, although it is larger in market share than all national brands except Charmin. Scott is affected by one of the premium brands (Soft 'N Pretty), and is marginally affected by Coronet (0.10 level). Scott itself affects all of the other moderate brands and marginally affects generics (0.10 level). Thus, the market is behaviorally treating Scott as a moderate brand, although under some physical volume comparisons with other brands, Scott may actually be a high-priced brand.

Chiffon margarine provides another insightful example. Chiffon is the lowest-priced premium national brand, with a regular price starting at 69¢ and falling to 59¢ (versus the other national brands at 83¢–89¢, and the private labels at 49¢–53¢). Thus, Chiffon is borderline premium, borderline moderate. Table 2 shows that Chiffon's prices affect none of the other premium brands, while it is affected by the prices of 3 of the other 4 premium brands. Chiffon marginally affects the Sunnyland moderate brand. Thus, it acts like an inferior premium, or a superior moderate brand, and occupies an intermediate tier of its own.

6.2. *Analysis of the Causes of and Patterns in Asymmetry*

Table 3 provides strong evidence consistent with at least bimodal preference distributions in the population. However, if there is also asymmetry in the absolute amount of deal price discounts empirically observed, then it is possible that the results are an estimation artifact of the historical pattern of dealing in the market. Analysis of the depth of deals show that the margarine and bathroom tissue categories have comparable deal depth ranges for the various types of brands. In the tuna and flour categories, the private label brands tend to have smaller deal discounts (in absolute pennies off and percentage terms) than the national brands in those categories. For tuna, the maximum deal depth for the private label is 15.2%, versus an average maximum deal depth for the national brands of 19.2%. For flour, the maximum private label deal discount is 10.1%, versus a national brand average maximum of 15.6%. Thus, it may be that if the private labels or generics increased their depth of deal in the tuna and flour categories, significant cross-price effects with the national brands might be found. However, the conclusion is still valid that at this lower but significant level of discounting, the relative preference distribution is consistent with at least bimodal preferences in order to find significant national brand effects on the private labels but not vice versa.

The specific pattern of asymmetries observed in Table 2 shows primarily upper-tier brands affecting other brands in their own tier and in the tier below, while lower-tier brands rarely affect upper-tier brands. This is the result of both bimodal preference distributions *and* the location of the current regular price indifference point fitting the general pattern shown in Figure 2a.

Table 4 presents another way of looking at the data in order to infer the general location of the regular price indifference point. The table concentrates on those *between-tier* brand comparisons which have estimable cross-price terms for both brands, excluding the generic brand comparisons. The brand-level focus of Table 4, as opposed to Table 3's price tier focus, allows a more precise determination of the location of the regular price indifference point. Since Chiffon and Scottissue are distinct tiers as determined above, Table 4 separates out these brands from the other national brands.

Brand comparisons are classified based upon the elasticity reported in Table 2. For example, since Pillsbury flour affects the private label Jewelmaid flour when it deals, but not vice versa, this comparison is classified as a "$N \rightarrow PL$"—i.e., the national (upper tier) brand affects the private label (lower tier), but not vice versa.

Table 4 reveals that in 32 of 34 cases, the strength of preference for quality distribution is consistent with bimodal preference distributions. In 15 cases, the regular price indifference point is located to the left of the distribution as shown in Figure 2a. This conclusion holds since reductions in the upper-tier brand's price affects the lower-tier brand's unit sales, but reductions in the lower-tier brand's price has no significant effect on the upper-tier brand's sales.

An additional 15 cases imply an indifference point around the middle of the trough between peaks in the quality distribution, as shown in Figure 2c. The brands are not price competitive for price deal decreases of the magnitudes in the data base.

In only 2 cases is there evidence that the private label brand affects the national brand but not vice versa. Thus, in 30 of 34 cases, the regular price indifference point is either in the center or toward the left of the trough (Figures 2c and 2a respectively), while in

TABLE 4

Summary of Brand Competition-Paired Comparisons

	N ↔ PL[1]	N → PL[2]	N ← PL[3]	N ↮ PL[4]
FLOUR	1	1		1
MARGARINE				
N-Chiffon		3		1
N-PL		4	1	3
Chiffon-PL		1		1
BATHROOM TISSUE				
N-Scott		1		3
N-PL		2	1	5
SCOTT-PL	1	1		
TUNA FISH	–	2	–	1
TOTALS	2	15	2	15

[1] National and Private Label Affect One Another.
[2] National Affects Private Label, Not Vice Versa.
[3] Private Label Affects National, Not Vice Versa.
[4] Neither Affects the Other.

only 2 cases do the results indicate an indifference point to the right as shown in Figure 2b.

In the remaining 2 cases, both brands affect one another. This is consistent with Figure 1 or Figure 2, where in the latter case the modes are closer together due to perceived quality $(q_i - q_k)$ differences being smaller than for the other national-private label comparisons.

7. Discussion, Implication and Conclusions

The preceding results allow several conclusions to be drawn about how price competition may influence observed market structure. Promotional price competition can logically result in asymmetric sales effects within a retail grocery category. For a majority of cases observed, higher-price-tier, higher-quality brands draw market share from their own price tier competitors, and typically from the tier below. However, the lower-quality, lower-price-tier brands take unit sales from own tier and tier below brands, but rarely take sales from higher-priced, higher-quality brands in the tier above. These results indicate that the preference distribution may be at least bimodal in the population, *and* that in this particular market the current regular price equilibrium is located either near the left side of the distribution as shown in Figure 2a or near the middle of the trough between modes as shown in Figure 2c. The fact that the structure of the asymmetry is so specific—lower tier brands generally do not significantly affect upper tier brands unless deal discounts are extreme—implies also that the market is treating the differing price tiers as significantly different in quality and hence the perceived quality difference $(q_i^c - q_k^c)$ is substantial enough to preserve the separation in the peaks of the quality θ^c distribution. The markets analyzed are basically homogeneous in product features except for quality and only the price levels and implied quality levels vary. Thus, there is at least some support for the Mussa and Rosen (1978) proposition that multiple quality levels for an otherwise undifferentiated product do in fact exist, and that manufacturers and retailers price these differing quality levels in such a way as to create price-quality tiers in the marketplace.

The asymmetries raise questions about the existing techniques' ability to recover market structures. Since competitive effects may be asymmetric, one brand may be a strong competitor of a second brand when dealing, but the second brand may have no effect on the first. The majority of market structure, perceptual mapping and conjoint techniques thus have problems depicting asymmetric types of price competition.[12] DEFENDER (Hauser and Shugan 1983) is a notable exception because it can allow differential competitive responses. While useful as a proxy for quality, price may not be a perfect correlate of quality *as perceived* in the market. The above research, especially in its derivation of competition from the underlying quality preference distribution, assumes an objective measure of quality q_i for each brand. In reality, heterogeneity in individual perceptions of the relative quality of the brands is likely to influence brand choice in the market. Thus, it is perhaps surprising that the asymmetries found in this research appear to be so clearly defined.[13]

One major question is why the regular price indifference point should so often be found at the left instead of right side of the relative preference distribution. In many markets in the United States, private label brands command greater profit margins than national brands. Hence, to have the national brands cannibalizing the private labels within a retail line may not seem optimal. We might also question why the retailer does not simultaneously raise the prices of all the national brands, thus shifting the regular price indifference point right. The equilibrium would then resemble Figure 2b. (Note that if all national brands are raised in price by an equal absolute-penny amount, the relative prices $(p_i - p_k)$ remain the same for all national brand comparisons.) The major reason why this may not occur in the market is that such changes can raise the regular price of the brands above the reservation price of some of the consumers, causing them to switch not just to other brands but to other lower-priced retailers to purchase a particular brand. In the Chicagoland market, Jewel Food Stores faces a host of price-competitive retailers, including both discounters and warehouse operations as well as comparably-priced chains.[14] In markets that allow higher national brand prices relative to private label brands (e.g. markets with less price competition among retailers), or that have higher-quality private label brands, the indifference point may be shifted rightward and result in private label brands having a more-significant impact on the premium brands.

8. Summary and Future Areas of Research

This paper developed a model for analyzing and explaining price-induced patterns of competition. Three major classes of distributions of the population's preference distribution were proposed to examine how price might affect competition. Based upon the basic utility theory and the various preference distributions, it is possible to predict what the market's competitive structure should look like. The empirical evidence argues that the preference distribution across different quality brands in the population is consistent with at least bimodal relative preference distribution (i.e. there may be "quality" and "economy" seekers in the population). In addition, the current empirical application generally shows a regular price equilibrium located toward the lower peak of this distribution, i.e. closer to the "economy" seekers portion of the distribution. This causes a

[12] In fairness, these techniques are generally most interested in depicting attribute competition—which competitors are close attribute substitutes of one another—and not which brands are actually competitive in the real world when prices are included. Cooper and Nakanishi (1988) present one procedure for attempting to build asymmetries into perceptual maps.

[13] One outside influence not accounted for (except in the empirical estimates of the base levels of sales) is how brand image advertising affects consumers' perception of quality. The highly advertised national brands tend to have substantially larger market shares than the less advertised national brands within the same tier. If advertising reinforces or heightens the quality image of a national brand, it can shift the regular price indifference point leftward. Thus the brand captures a larger proportion of the consumer's relative preference distribution.

[14] Of course, one additional reason why such pricing behavior may occur is that buying and to a degree retail pricing decisions are made contemporaneously by factoring in the forward buy savings when brands are bought at reduced cost on price deals from manufacturers, and then are sold later at the regular (hence substantially higher-margin) retail price.

very specific competitive structure to occur. When higher-price, higher-quality brands price deal, they steal unit sales away from other brands in their own price tier and from brands in the tier below (the moderate and private label brands). However, when lower-price, lower-quality brands deal, they draw sales from their own tier (other moderate and private label brands) and the tier below (generics), but in general do *not* take significant amounts of unit sales away from the tier above (national brands).

The specific results apply to the historical Chicagoland market. However, it is possible that the bimodal relative preference distribution holds across geographic regions. Thus, methods to determine market competition should take into account possible asymmetries in brand competition.[15] It is less clear that the regular price equilibria locations, hence the indifference points I_{ik}, will remain constant across markets. Local competitive circumstances may allow an upward shift in the price of the national brands, shifting the regular price indifference point to the right and making private labels less exposed to price promotion competition from the national brands. Future work on the supply side would also be of value in generating and equilibrium model. Further work is also required on the conditions necessary for profits to be optimized and the implication of these conditions for regular price equilibria in the market. This includes a better understanding of how to set retail and manufacturer regular and promotional price levels to maximize retailer and manufacturer profits.

Finally, this research concentrated on price-induced patterns of competition, and thus analyzed brands whose product forms, attributes, and sizes were comparable. There may also be significant own-brand size cannibalization when only one size of a given brand price deals. Thus, if the 12-ounce size deals, it may cannibalize sales of the 6-ounce and 18-ounce sizes of the same brand, due to cost-per-ounce factors for an identical (same brand formulation) product substitute. Additionally, size loyalties may exist in some markets, making switching between comparably sized brands much more likely than switching to other sizes of a competitive brand. Further work is needed to see what degree of form loyalties exist (e.g. tub versus stick margarine buyers, oil versus water tuna buyers, etc.), and/or if such loyalties can be related to price differentials caused by differing ingredients or production technologies.[16]

Acknowledgments. This research was partially funded by the Center for Research in Marketing, by the Graduate School of Business of the University of Chicago, and by the National Science Foundation Grant SES 8421165. Our thanks are also extended to the Jewel Food Store Company and to Tom Richardson for the provision of the scanning data, to Linus Schrage, Chakravarthi Narasimhan, Ward Hanson, Steve Shugan, J. Brad Barbeau, Greg Allenby, and Sridhar Moorthy for helpful comments. The Area Editor and three anonymous reviewers also contributed valuable comments. Authors are listed alphabetically.

[15] Carpenter et al. (1988) in an article published subsequent to the research reported here propose an asymmetric attraction model for modeling asymmetric competition. Their empirical example from an Australia product category also exhibits competition of the form that lower-priced economy brands compete primarily with themselves, but have little influence on other higher-priced, higher-quality brands. Russell and Bolton (1988) also find a much greater tendency for private label brands to switch across subgroups versus national brands.

[16] This paper was received in December 1986 and has been with the authors 15 months for 3 revisions.
Accepted by Subrata K. Sen.

Bibliography

Allenby, Gregory M. (1989), "A Unified Approach to Identifying, Estimating and Testing Demand Structures with Aggregate Scanner Data," Working Paper, College of Business, Ohio State University.

Batsell, Richard G. and John C. Polking (1985), "A New Class of Market Share Models," *Marketing Science*, 4, 3 (Summer), 177–188.

Blattberg, Robert C. and Kenneth J. Wisniewski (1987), "How Retail Price Promotions Work: Empirical Results," Center for Research in Marketing, Working Paper #42, University of Chicago, (December).

——— and ——— (1988), "Modeling Store Level Scanner Data," University of Chicago, Marketing Working Paper #43, (January).

Carpenter, Gregory S., Lee G. Cooper, Dominque M. Hanssens and David F. Midgley (1988), "Modeling Asymmetric Competition," *Marketing Science*, 7, 4 (Fall), 393–412.

Cattin, Philippe and Dick R. Wittink (1982), "Commercial Use of Conjoint Analysis: A Survey," *Journal of Marketing*, 46, 3 (Summer), 44–53.
Cooper, Russel (1984), "On Allocative Distortions in Problems of Self-Selection," *Rand Journal of Economics*, 15, 4 (Winter), 568–577.
Cooper, Lee G. and Masao Nakanishi (1988), *Market Share Analysis*, Boston: Kluwer Academic Publishers.
Eskin, Gerald J. and Penny Baron (1977), "Effects of Price and Advertising in Test Market Experiments," *Journal of Marketing Research*, 14 (November), 499–508.
Gabor, Andre and C. W. J. Granger (1966), "Price as an Indicator of Quality: Report on an Enquiry," *Economica*, 33, 129 (February), 43–70.
Green, Paul E. (1973), *Multiattribute Decisions in Marketing*, Hindsdale, IL: The Dryden Press.
——— and Yoram Wind (1975), "New Way to Measure Consumer's Judgments," *Harvard Business Review*, (July-August), 107–117.
——— and V. Srinivasan (1978), "Conjoint Analysis in Consumer Research—Issues and Outlook," *Journal of Consumer Research*, 5, 2 (September), 103–123.
Guadagni, Peter M. and John D. C. Little (1983), "A Logit Model of Brand Choice Calibrated on Scanner Data," *Marketing Science*, 2, 3 (Summer), 203–238.
Hauser, John R. and Steven P. Gaskin (1983), "Application of the 'DEFENDER' Consumer-Model," *Marketing Science*.
——— and Steven M. Shugan (1983), "Defensive Marketing Strategies," *Marketing Science*, 2, 4 (Fall), 319–360.
——— and Patricia Simmie (1981), "Profit Maximizing Perceptual Positions," *Management Science*, 27, 1 (January), 33–56.
Johnson, Richard M. (1974), "Trade-Off Analysis of Consumer Values," *Journal of Marketing Research*, 11 (May), 121–127.
Jones, J. M. and Fred S. Zufryden (1980), "Adding Explanatory Variables to a Consumer Purchase Behavior Model: An Exploratory Study," *Journal of Marketing Research*, 17 (August), 323–334.
Kalwani, Manohar and Donald G. Morrison (1977), "A Parsimonious Description of the Hendry System," *Management Science*, 23, 467–477.
Kinberg, Yoram, Ambar G. Rao and Melvin F. Shakon (1974), "A Mathematical Model for Price Promotions," *Management Science*, 20, 6 (February), 948–959.
Lilien, Gary L. and Philip Kotler (1983), *Marketing Decision Making—A Model Building Approach*, New York: Harper and Row.
McAlister, Leigh (1985), "The Impact of Price Promotions on a Brand's Market Share, Sales Patterns, and Profitability," MIT Working Paper 1622-85, (February).
Moorthy, K. Sridhar (1984), "Market Segmentation, Self-Selection, and Product Line Design," *Marketing Science*, 3, 4 (Fall), 288–307.
Mussa, Michael and Sherwin Rosen (1978), "Monopoly and Product Quality," *Journal of Economic Theory*, 18, 301–317.
Oren, Shmuel, Stephen Smith and Robert Wilson (1984), "Product Line Pricing," *Journal of Business*, 57, 1, 2, S73–S99.
Rao, Vithala R. (1984), "Pricing Research in Marketing: The State of the Art," *Journal of Business*, 57, 1, 2 (January), S39–S64.
Reibstein, David J. and Hubert Gatignon (1984), "Optimal Product Line Pricing: The Influence of Elasticities and Cross-Elasticities," *Journal of Marketing Research*, 21 (August), 259–267.
Russell, Gary J. and Ruth N. Bolton (1988), "Implications of Market Structure for Elasticity Structure," *Journal of Marketing Research*, 25 (August), 229–241.
Shugan, Steven M. (1987), "Estimating Brand Positioning Maps Using Supermarket Scanning Data," *Journal of Marketing Research*, 24, 1 (February), 1–18.
Sunoo, Don and Y. S. Lin (1978), "Sales Effects of Promotion and Advertising," *Journal of Advertising Research*, 18, 5 (October), 37–40.
Thaler, Richard (1985), "Mental Accounting and Consumer Choice," *Marketing Science*, 4, 3 (Summer), 199–214.
Urban, Glen L. and John R. Hauser (1980), *Design and Marketing of New Products*, Englewood-Cliffs, NJ: Prentice-Hall, Inc.
———, Philip L. Johnson and John R. Hauser (1984), "Testing Competitive Market Structures," *Marketing Science*, 3, 2 (Spring), 83–112.
Wilkinson, J. B., J. Barry Mason and C. H. Paksoy (1982), "Assessing the Impact of Short-Term Supermarket Strategy Variables," *Journal of Marketing Research*, 19 (February), 72–86.
Wisniewski, Kenneth J. and Robert C. Blattberg (1988), "Analysis of Consumer Response to Retail Price Dealing Strategies: A UPC-Based Approach For Modeling Deal Sales Response and Optimizing Deal Profitability," National Science Foundation Final Report Grant SES 8421165, (April).